



A multi-layer retrieval-augmented large language model framework for enhancing hypertension education

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Abstract

Large Language Models (LLMs) demonstrate considerable potential in enhancing the retrieval of health information. However, the hallucinatory they produce poses a security challenge. This study aimed to improve the accuracy and reliability of LLMs in hypertension education through the integration of Retrieval-Augmented Generation (RAG) technology. We constructed a hypertension supplement knowledge base, and subsequently integrated it into a RAG technology, resulting in the development of the HEART (Hypertension Enhancing Answer Retrieval Tool) framework. A set of 50 commonly asked questions related to hypertension was used to evaluate the performance of four base LLMs—ChatGPT-4o, Claude-3.5, Gemini-2.5, and Llama-3.3—as well as their corresponding HEART-enhanced versions. Clinical experts assessed each response in terms of accuracy, completeness, consistency, robustness, security, and overall quality. The integration with the HEART framework led to a significant improvement in the performance of all four LLMs across five key evaluation dimensions: accuracy, completeness, consistency, security, and robustness (all $P < 0.05$). The average overall quality scores for all models increased significantly: from 3.57 (SD 0.72) to 4.20 (SD 0.41) for Llama-3.3, from 3.92 (SD 0.70) to 4.38 (SD 0.42) for Claude-3.5, from 3.91 (SD 0.73) to 4.32 (SD 0.39) for ChatGPT-4o, and from 4.03 (SD 0.69) to 4.38 (SD 0.41) for Gemini-2.5 (all $P < 0.001$). This study highlights the importance of combining high-quality, domain-specific medical data with advanced artificial intelligence techniques to enhance accuracy and reduce misinformation in healthcare applications.

Keywords Digital hypertension · Health Literacy · Implemental hypertension · Large Language Models · Patient Education · Retrieval-Augmented Generation

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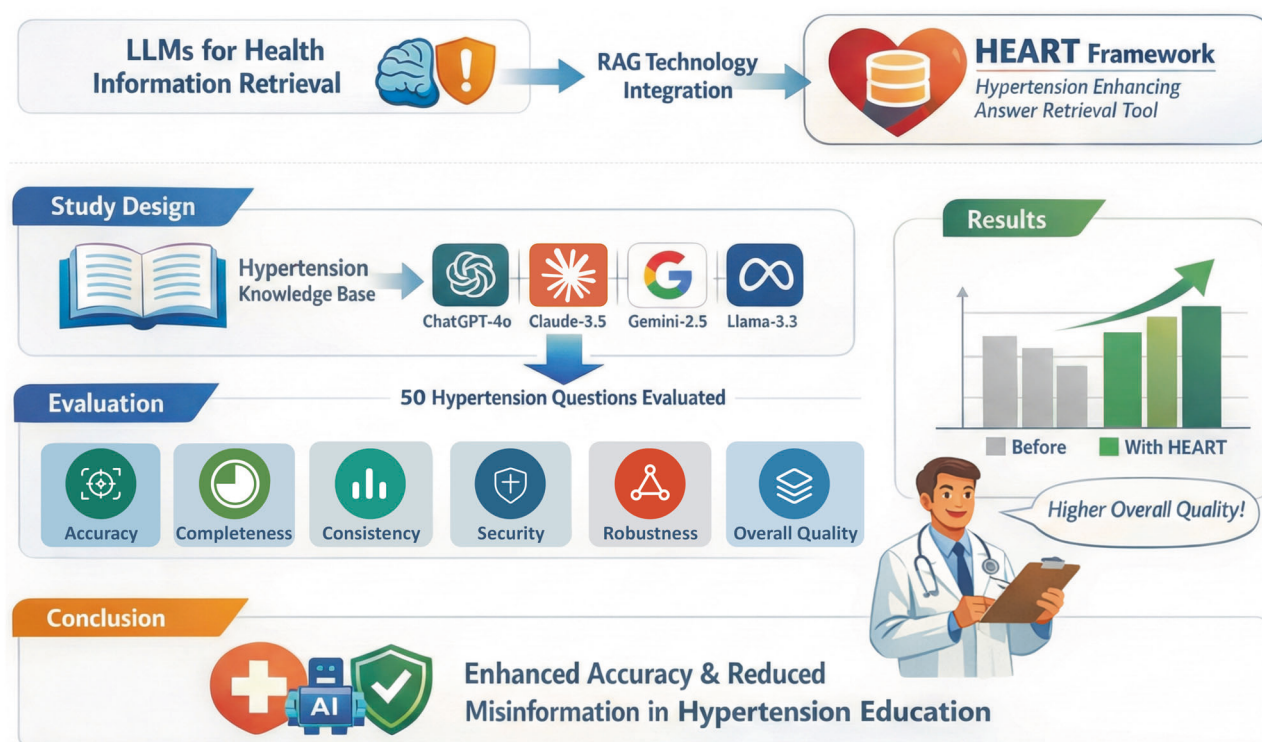
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Introduction

Hypertension represents one of the most prevalent cardiovascular conditions globally, affecting over 1.3 billion individuals worldwide and posing substantial challenges for healthcare systems [1–4]. The complexity of hypertension management—encompassing diagnostic criteria, therapeutic interventions, lifestyle modifications, and monitoring protocols—demands highly reliable information systems capable of providing clinically sound guidance [5, 6]. However, the quality and accessibility of evidence-based hypertension information remain suboptimal across healthcare platforms globally [7–9]. This challenge is exemplified in populous countries such as China, where hypertension prevalence has exceeded 250 million individuals while effective control rates remain inadequate [9]. A comprehensive analysis revealed that 78.5% of hypertension-related information on major digital platforms received poor credibility ratings, with 75.4% of content lacking essential treatment guidance consistent with established clinical

Graphical Abstract



guidelines [10]. These findings reflect a broader international problem wherein widespread availability of online health resources does not guarantee accuracy or clinical utility. Consequently, there exists an urgent global need for accessible, trustworthy, and guideline-consistent information delivery systems to support patient education and clinical decision-making.

Large Language Models (LLMs) have revolutionized natural language processing capabilities, demonstrating unprecedented performance across diverse applications, including conversational agents, automated content generation, and information retrieval systems [11–14]. These sophisticated neural architectures have exhibited remarkable proficiency in understanding complex linguistic patterns and generating contextually relevant responses, positioning them as transformative tools in numerous domains requiring human-computer interaction [15]. Despite these advances, LLMs remain susceptible to a critical limitation known as hallucination—the generation of plausible yet factually incorrect or unsupported information [16]. This phenomenon represents a fundamental challenge wherein models produce outputs that appear coherent and authoritative but lack grounding in verifiable data or established knowledge [17]. The implications of such fabricated content become particularly concerning in high-stakes domains where accuracy is paramount for safety and decision-making [18].

Retrieval-Augmented Generation (RAG) has emerged as a promising approach for enhancing LLMs reliability by integrating external knowledge sources with generative

capabilities. RAG frameworks enable models to access and incorporate relevant information from curated databases during the response generation process, potentially reducing hallucination rates while improving factual accuracy [19]. However, the effectiveness of RAG implementations is critically dependent on the quality, completeness, and domain-specificity of the underlying knowledge repositories. The effectiveness of RAG implementations remains critically dependent on the quality, completeness, and domain-specificity of underlying knowledge repositories.

In this study, we aimed to evaluate the efficacy of a novel RAG-based framework, named the HEART (Hypertension Enhancing Answer Retrieval Tool), in enhancing the accuracy, completeness, and safety of responses generated by base LLMs when answering hypertension-related questions.

Methods

Hypertension supplement knowledge base

To provide domain-specific knowledge for the HEART framework, we constructed and iteratively refined Hypertension Supplement Knowledge Base 2.0 (HSKB2.0). This knowledge base was constructed by compiling a comprehensive corpus of hypertension-related content derived from clinical practice guidelines and authoritative medical

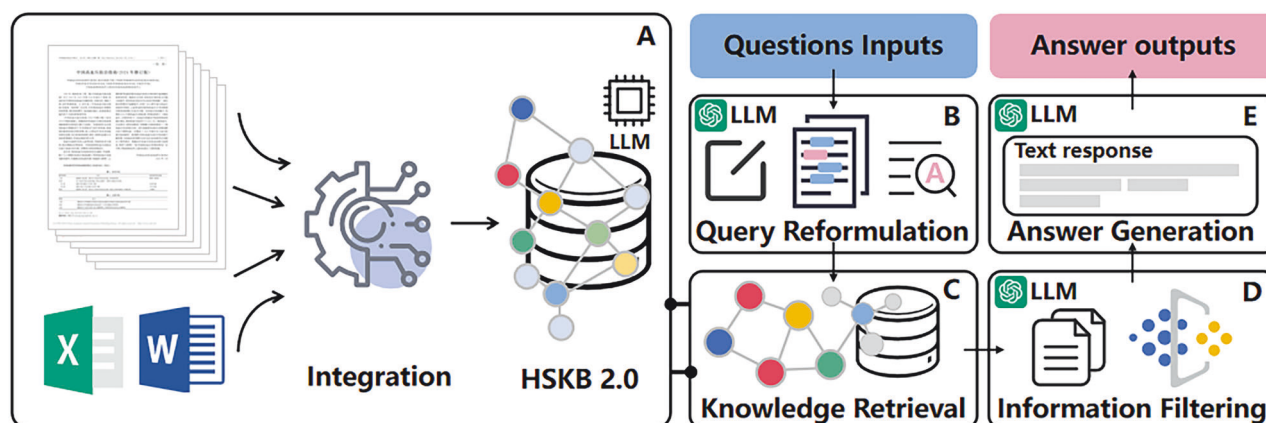


Fig. 1 The overall design of the HEART (Hypertension Enhancing Answer Retrieval Tool) framework. **A** HSKB 2.0 construction; **B** query reformulation; **C** knowledge retrieval; **D** information filtering; **E**

answer generation. LLM large language model, HSKB hypertension supplement knowledge base

knowledge platforms (Fig. 1) [20, 21]. Key sources included the *Chinese Hypertension Prevention and Treatment Guidelines (2024 Revision)*, *Cardiovascular Medicine of Sports and Exercise Guidelines (2024)*, *Chinese Guidelines for the Management of Hypertension in the Elderly (2023)*, *Adult Hypertension Dietary Guidance (2023)*, *National clinical practice guidelines on the management of hypertension in primary health care in China(2020)*, *National Guideline for Hypertension Management in China (2019)*, *the Chinese Clinical Guidelines for Smoking Cessation (2015)*. Additionally, it incorporated a curated set of over 1500 standardized question-answer pairs, which were collected from the web and rigorously screened and validated by medical experts. To ensure their clinical relevance and accuracy, these pairs underwent a thorough validation process. Initially, a draft set of over 1500 questions was formulated based on established hypertension guidelines and literature. This set was then subjected to multiple rounds of review and refinement through expert discussions, which included clinicians and specialists in hypertension management. The iterative review process allowed for the identification and correction of any discrepancies, ensuring that the questions and their corresponding answers were both evidence-based and aligned with current best practices in hypertension management. This collaborative expert-led process helped validate the content and ensure its applicability for the HEART framework. The database was designed to comprehensively cover all major domains of hypertension care, including pathophysiology, diagnosis, treatment, long-term management, and patient education, while ensuring that none of the questions used in the evaluation phase were directly included.

All retrieved documents underwent rigorous preprocessing to eliminate unstructured or noisy content such as figures, tables, references, and author disclosures. After

cleaning, documents were segmented using the Parent-Child Chunk Mode, a two-layer chunking approach: Parent chunks were created for contextual understanding (mode: “paragraph”, delimiter: “##”, max length: 2048 tokens); Child chunks were created for fine-grained retrieval (delimiter: “/n”, max length: 1024 tokens). Next, the segmented content was indexed using the high-quality index method with Qwen’s “text-embedding-v3” as the embedding model. Retrieval was configured using hybrid search, which combines full-text keyword matching with semantic vector search. A re-ranking mechanism was implemented, with 70% weight assigned to keyword relevance and 30% to semantic similarity, to ensure the selection of the most relevant document fragments.

Retrieval-augmented framework

To improve the performance of LLMs in hypertension-related question answering, we developed the HEART framework based on an RAG architecture. This framework consists of four layers: Query Reformulation, Knowledge Retrieval, Information Filtering, and Answer Generation (Fig. 1).

Query reformulation

The initial layer involves reformulating the user’s query to improve retrieval performance. Each query is processed through a base LLM (ChatGPT-4o, Claude-3.5, Gemini-2.5, or Llama-3.3) to enhance clarity and expand semantic coverage. This process includes correcting typographical errors, resolving abbreviations, and incorporating synonyms or related terms to maximize the retrieval of relevant knowledge segments. LLMs rewrite queries using the following prompt:

“You are a medical assistant helping to rewrite user queries to improve information retrieval. Please rewrite the following query to make it clearer and more complete. Correct any potential spelling or typographical errors, and consider using synonyms or expanded terms if appropriate, while preserving the original meaning”.

Knowledge retrieval

The processed query was used to retrieve relevant information from HSKB2.0. The query was embedded using the Qwen’s “text-embedding-v3” model, and retrieval was conducted using Hybrid Search, which integrates full-text keyword matching and dense vector similarity. The resulting documents were re-ranked with weighted scores (70% keyword relevance, 30% semantic similarity), and the top $k = 10$ document chunks with a combined weighted score greater than 0.3 are selected as contextual knowledge to guide response generation.

Information filtering

To ensure the reliability and appropriateness of generated responses, the HEART framework incorporates a structured safety filtering mechanism designed to mitigate misinformation, harmful advice, and inappropriate medical suggestions. This mechanism is governed by 8 predefined safety principles, which are explicitly embedded into the model’s prompting instructions as follows: *“You are an experienced cardiovascular physician in China. You need to filter the retrieved background content according to the following rules:*

1. *Avoid personal diagnosis or treatment suggestions*
2. *Prohibit recommendations of specific prescription drugs*
3. *Discourage use of non-evidence-based therapies*
4. *Emphasize seeking professional medical care when necessary*
5. *Avoid speculative, exaggerated, or overly definitive clinical statements*
6. *Prohibit endorsement of commercial health products or services*
7. *Avoid providing misleading or outdated health advice*
8. *Clearly indicate uncertainty when evidence is lacking or ambiguous”*

Answer generation

Finally, the validated and filtered information was fed back into the base LLMs along with a structured prompt to generate the final response. The prompt instructs *“You are an experienced cardiovascular physician in China. Using*

the verified and filtered context below, please provide a concise and accurate response to the user’s query

Keep the response under 300 words.

Ensure all information is accurate and consistent with the provided context.

If the answer is unknown or cannot be determined from the context, explicitly state “I do not know” rather than fabricating information.

Maintain a neutral, professional tone throughout your response”.

Performance evaluation

Overview

This evaluation was conducted from April 23 to May 20, 2025 at West China Hospital of Sichuan University (Chengdu, China). A total of 50 hypertension-related questions were selected based on a prior study involving hypertensive patient-LLMs interactions (Table 1) [20]. These questions, covering four key domains-blood pressure knowledge, blood pressure measurement, blood pressure management, and medication management-were designed to reflect the most frequently asked topics and common informational needs of patients and the general public regarding hypertension care.

Respond Generation

To assess model performance, we submitted a set of 50 hypertension-related queries to eight models-four base LLMs (ChatGPT-4o, Claude-3.5, Gemini-2.5, and Llama-3.3) and their corresponding HEART-enhanced versions. A detailed parameter and source description of the base LLMs can be found in Supplementary Table 1. Each question was treated as an independent input, with prior conversation history cleared to avoid contextual bias. All model outputs were generated on iPad (Apple, America) and converted into plain text to eliminate identifying features [22, 23]. Responses were then evaluated in a blinded and randomized manner by three experienced clinicians from West China Hospital of Sichuan University and First Affiliated Hospital of Bengbu Medical University. Evaluations were conducted over eight rounds, with 48-hour intervals between rounds to reduce fatigue and potential carryover effects (Fig. 2).

Inter-rater reliability analysis

The inter-rater reliability analysis was conducted using the Intraclass Correlation Coefficient (ICC) to assess the agreement among the three clinical evaluators. The ICC

Table 1 English translated version of the hypertension-related question set used for large language model evaluation

Blood Pressure Knowledge	
1	What are systolic and diastolic blood pressure?
2	What is considered a normal blood pressure value?
3	How is hypertension diagnosed?
4	What are the clinical classifications (grades) of hypertension?
5	What does “high-normal blood pressure” mean in a health check-up report?
6	What examinations are recommended for hypertension evaluation?
7	What is primary (essential) hypertension?
8	What are the common symptoms of hypertension?
9	What are the major risk factors for developing hypertension?
10	What are the potential health complications of hypertension?
11	What is the target blood pressure for hypertension control?
12	Does blood pressure fluctuation affect health?
13	What is secondary hypertension?
14	What symptoms may indicate possible secondary hypertension?
15	Is isolated diastolic hypertension considered hypertension?
Blood Pressure Measurement	
16	Which method is most accurate: home BP monitoring, office BP, or ambulatory BP monitoring?
17	Why is my blood pressure higher when measured at the hospital?
18	Should I use an electronic or mercury sphygmomanometer?
19	How should I choose the correct cuff size for blood pressure measurement?
20	What is the best time of day to measure blood pressure?
21	How often should blood pressure be measured?
22	What is the correct method for measuring blood pressure?
23	Should blood pressure be measured on the left or right arm?
24	How many times should BP be measured per session, and when should it be repeated?
25	My BP readings vary with each measurement—which one is most accurate?
Blood Pressure Management	
26	Which vegetables may help lower blood pressure?
27	Which fruits are suitable for individuals with hypertension?
28	What types of meat are appropriate for hypertensive patients?
29	Do seafood products have antihypertensive effects?
30	What is the recommended daily salt intake for patients with hypertension?
31	What foods should hypertensive individuals avoid or limit?
32	What are the alcohol consumption limits for hypertensive patients?
33	What is the ideal weight range to support blood pressure control?
34	What types of physical activity are recommended for people with hypertension?
35	What is the appropriate intensity and frequency of exercise for hypertension management?

Table 1 (continued)

Blood Pressure Knowledge	
36	Can patients with hypertension use hot springs or saunas?
37	How can psychological stress-induced blood pressure elevations be managed?
38	How can poor sleep-related blood pressure increases be addressed?
39	Can switching from cigarettes to e-cigarettes help reduce blood pressure?
Medication Management	
40	Is medication always necessary after being diagnosed with hypertension?
41	Does long-term use of antihypertensive drugs lead to tolerance?
42	Can prolonged use of antihypertensive drugs damage the liver or kidneys?
43	What should I do if my blood pressure remains high despite taking medication?
44	Should antihypertensive drugs be taken in the morning or evening?
45	Should antihypertensive medication be withheld before a medical appointment?
46	Is treatment required for isolated diastolic hypertension?
47	Should multiple antihypertensive medications be taken separately?
48	Are fixed-dose combination drugs more effective for lowering blood pressure?
49	Which commonly used medications should not be combined with antihypertensive drugs?
50	What are the main types of medications used to treat hypertension?

analysis was carried out using a two-way random effects model with absolute agreement (ICC(A,3)), which is deemed the most suitable approach for the current study design, where multiple raters evaluate the same responses across several assessment dimensions.

Model evaluation

Model evaluations were conducted across five key dimensions: accuracy (factual correctness), completeness (coverage of all relevant aspects of the query), consistency (stability of responses when the same question was submitted twice), security (potential for causing harm or misleading patients), and robustness (resilience to input disturbances such as typos, grammatical errors, or reordered phrasing) (Supplementary Table 2. For consistency assessment, each query was submitted twice under identical conditions, and the outputs were compared for semantic and clinical coherence. For robustness assessment, intentionally perturbed versions of the original questions were used, and the resulting responses were compared with those from the original inputs to evaluate quality preservation.

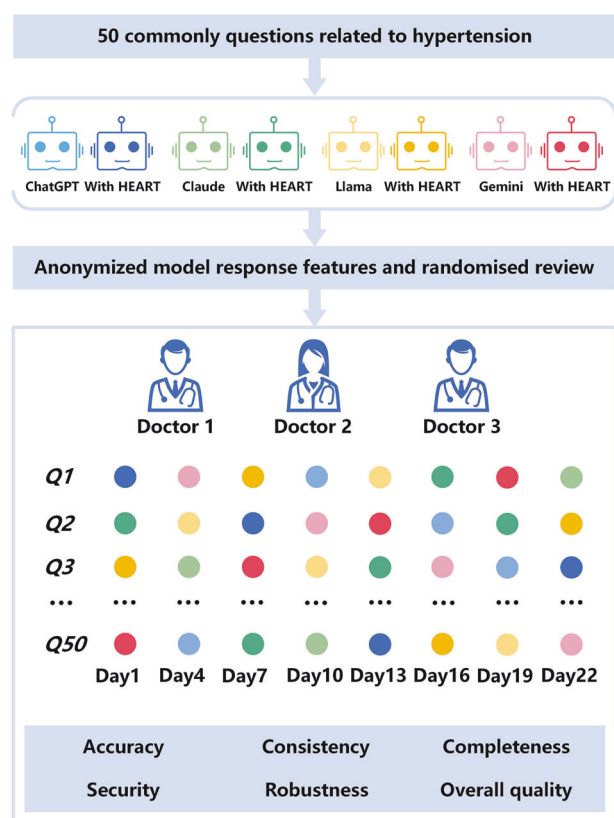


Fig. 2 Flowchart of the overall study design. This study evaluates the performance of four publicly available large language models and their corresponding HEART-enhanced versions in responding to common hypertension-related questions. Evaluation is conducted by clinicians across six key dimensions: accuracy, completeness, consistency, security, robustness, and overall response quality

Supplementary Table 3 lists the 50 hypertension-related questions and their robustness-test variants that were submitted to both base and HEART-enhanced LLMs for evaluation.

Each response was independently rated on all five dimensions using a 5-point Likert scale, where 1 indicated very poor performance and 5 indicated excellent performance. Scores below 3 were flagged as “bad responses,” indicating that the output was inadequate or potentially misleading in that particular dimension.

To assess the overall quality of each response, we computed the arithmetic mean of its scores across the five dimensions. Responses with an average score below 3.0 were considered suboptimal and potentially unsafe for clinical use.

Statistical analysis

Statistical analyses were performed using SPSS (version 23.0; IBM Corp.) and R (version 4.2.2). The normality of continuous variables was assessed using the Kolmogorov–Smirnov test. Variables with a normal distribution were expressed as mean $\pm$ standard deviation (SD) and compared using

independent-sample *t* tests. Non-normally distributed variables were reported as medians and interquartile ranges (IQR) and analyzed using the Mann–Whitney *U* test (Wilcoxon rank sum test). Categorical variables were summarized as counts and percentages, and differences between groups were assessed using the chi-square test. A two-tailed *P* value < 0.05 was considered statistically significant.

Ethical Considerations

This study was approved by the Ethics Committee of the First Affiliated Hospital of Bengbu Medical University (Approval No. [2024]KY066). The trial was conducted in accordance with clinical practice guidelines, the Declaration of Helsinki, and all relevant laws and regulations in China.

Result

Intraclass correlation coefficient

Our analysis revealed excellent inter-rater reliability with an ICC(A,3) value of 0.825 (95% CI: 0.809–0.839), indicating strong agreement among the three clinical evaluators according to Cicchetti’s criteria (0.75–1.00 = excellent reliability). The statistical significance was confirmed through F-test analysis ($F(1999,2481) = 5.79$, $P < 0.001$), providing strong evidence against the null hypothesis of zero correlation. The narrow confidence interval width (0.030) demonstrates high precision in our reliability estimate, with values consistently exceeding the acceptable threshold for clinical research applications.

Accuracy evaluation

We evaluated the accuracy of four base LLMs and their HEART-enhanced counterparts in responding to hypertension-related queries. As shown in Fig. 3 and Supplementary Table 4, the average accuracy scores of all four base models improved substantially following integration with the HEART framework. For Llama-3.3, the mean accuracy score increased from 3.46 (SD 0.92) to 4.19 (SD 0.73) ($P < 0.001$, Cohen’s $d = 0.879$). Claude-3.5 exhibited a similar pattern, with its average score rising from 3.87 (SD 0.91) to 4.36 (SD 0.53) ($P < 0.001$, Cohen’s $d = 0.658$). Gemini-2.5 showed a modest but statistically significant improvement, with the score increasing from 4.12 (SD 0.81) to 4.37 (SD 0.55) ($P = 0.011$, Cohen’s $d = 0.361$). Finally, for ChatGPT-4o, the mean accuracy score improved from 3.99 (SD 0.77) to 4.30 (SD 0.50) ($P = 0.005$, Cohen’s $d = 0.478$), accompanied by a noticeable increase in the number of high-scoring responses (Supplementary Table 4).

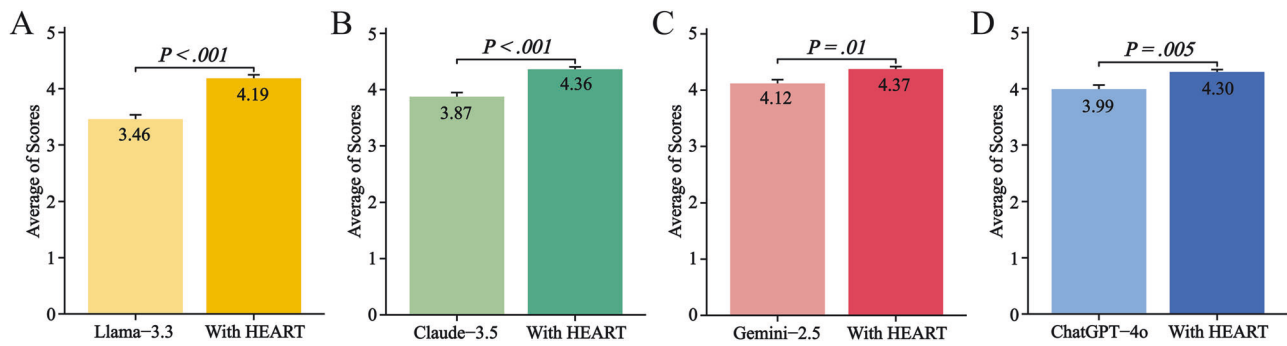


Fig. 3 Average scores for accuracy from large language models. **A** Llama-3.3; **B** Claude-3.5; **C** Gemini-2.5; **D** ChatGPT-4o. HEART Hypertension Enhancing Answer Retrieval Tool

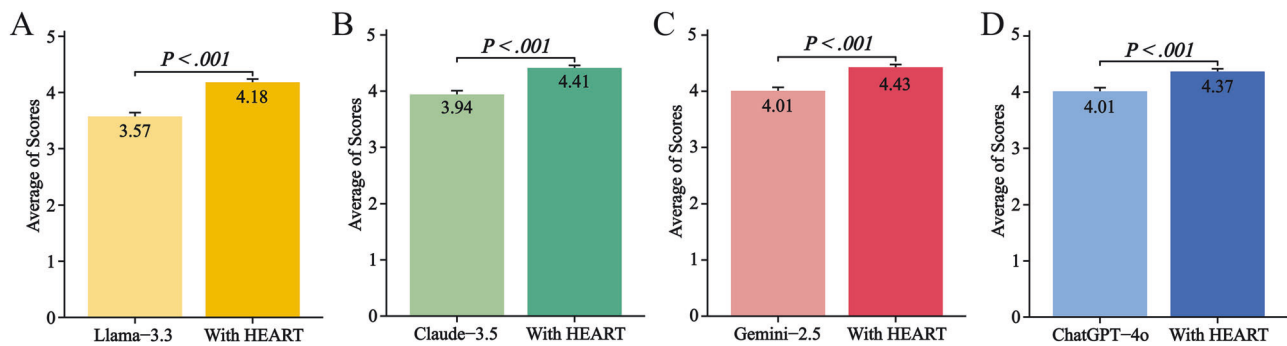


Fig. 4 Average scores for completeness from large language models. **A** Llama-3.3; **B** Claude-3.5; **C** Gemini-2.5; **D** ChatGPT-4o. HEART Hypertension Enhancing Answer Retrieval Tool

Completeness evaluation

We further evaluated the models' ability to provide complete and informative responses to hypertension-related questions. As shown in Fig. 4, all four LLMs achieved significant improvements in completeness after being enhanced with the HEART framework. For Llama-3.3, the average completeness score increased from 3.57 (SD 0.83) to 4.18 (SD 0.72) ($P < 0.001$, Cohen's $d = 0.785$), with the distribution shifting toward higher scores (Supplementary Table 5). Claude-3.5 showed similar improvement, with the score rising from 3.94 (SD 0.82) to 4.41 (SD 0.55) ($P < 0.001$, Cohen's $d = 0.673$). For Gemini-2.5, the completeness score rose from 4.01 (SD 0.76) to 4.43 (SD 0.59) ($P < 0.001$, Cohen's $d = 0.671$). ChatGPT-4o improved from 4.01 (SD 0.80) to 4.37 (SD 0.60) ($P < 0.001$, Cohen's $d = 0.509$), and its response distribution also demonstrated a marked reduction in partially complete or incomplete answers (Supplementary Table 5).

Consistency evaluation

To evaluate response stability, each question was submitted twice under identical conditions, and we assessed the degree of semantic and clinical coherence between the two outputs.

As shown in Fig. 5 and Table 2, all four models demonstrated significantly improved consistency after incorporating the HEART framework. For Llama-3.3, the average consistency score increased from 3.68 (SD 0.94) to 4.32 (SD 0.51) ($P < 0.001$, Cohen's $d = 0.846$), indicating a substantial improvement in output stability. Claude-3.5 exhibited a marked enhancement as well, with scores rising from 4.01 (SD 0.81) to 4.37 (SD 0.54) ($P < 0.001$, Cohen's $d = 0.523$). Gemini-2.5 improved from 4.00 (SD 0.78) to 4.32 (SD 0.53) ($P < 0.001$, Cohen's $d = 0.480$). For ChatGPT-4o, the average score increased from 3.75 (SD 0.85) to 4.39 (SD 0.55) ($P < 0.001$, Cohen's $d = 0.894$), suggesting a significant reduction in variability between repeated responses (Supplementary Table 6).

Security evaluation

We evaluated the safety of model outputs by assessing the potential for harmful, misleading, or clinically inappropriate content. As shown in Fig. 6 and Table 2, the HEART framework significantly improved the security scores across all four LLMs. For Llama-3.3, the average security score increased from 3.85 (SD 0.90) to 4.31 (SD 0.64) ($P < 0.001$, Cohen's $d = 0.589$). Claude-3.5 also exhibited notable improvement, with scores rising from 4.01 (SD 0.88) to

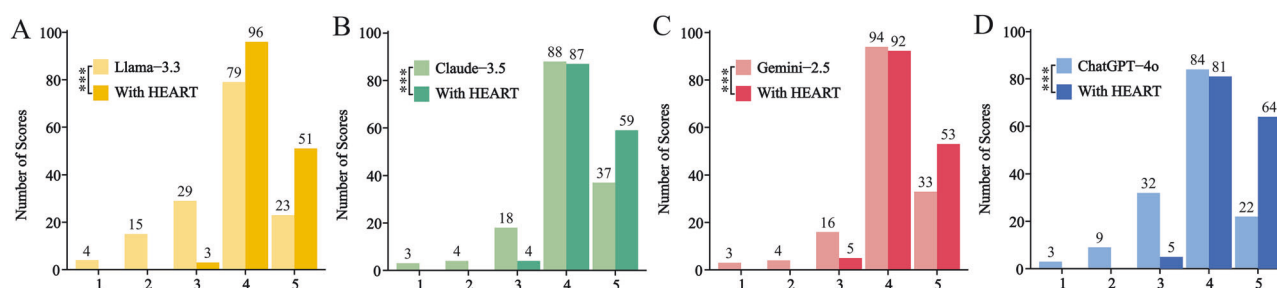


Fig. 5 Consistency score distributions of responses from large language models. **A** Llama-3.3; **B** Claude-3.5; **C** Gemini-2.5; **D** ChatGPT-4o. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$; HEART Hypertension Enhancing Answer Retrieval Tool

4.39 (SD 0.52) ($P < 0.001$, Cohen's $d = 0.526$). Gemini-2.5 achieved high baseline performance and further improved from 4.13 (SD 0.80) to 4.43 (SD 0.50) ($P = 0.010$, Cohen's $d = 0.450$) following HEART enhancement. For ChatGPT-4o, the average score improved from 4.06 (SD 0.87) to 4.36 (SD 0.50) ($P = 0.006$, Cohen's $d = 0.423$), with a shift in the distribution toward safer responses (Supplementary Table 7).

Robustness evaluation

To assess robustness, we introduced perturbed versions of each query—such as misspellings or reordered phrasing—and compared the model responses with those from the original queries. As shown in Fig. 7 and Supplementary Table 8, all four models showed significant gains in robustness after being enhanced with the HEART framework. For Llama-3.3, the average robustness score increased from 3.28 (SD 1.36) to 4.04 (SD 0.90) ($P < 0.001$, Cohen's $d = 0.659$), indicating improved resilience to input distortions. Claude-3.5 demonstrated similar improvements, with scores rising from 3.74 (SD 0.99) to 4.37 (SD 0.56) ($P < 0.001$, Cohen's $d = 0.783$), reflecting better semantic interpretation under noisy input conditions. Gemini-2.5 also showed a statistically significant improvement from 3.91 (SD 0.88) to 4.35 (SD 0.58) ($P < 0.001$, Cohen's $d = 0.590$). ChatGPT-4o improved from 3.73 (SD 0.90) to 4.20 (SD 0.72) ($P < 0.001$, Cohen's $d = 0.577$), showing fewer errors in handling typos or unstructured input patterns.

Overall quality evaluation

We evaluated overall model performance by calculating the arithmetic mean of each response's five-dimensional scores—accuracy, completeness, consistency, security, and robustness. As shown in Fig. 8A and Supplementary Table 9, all HEART-enhanced models exhibited a clear rightward shift in distribution compared to their corresponding base versions, indicating notable improvements in multidimensional response quality.

To further compare models, pairwise statistical tests across all eight configurations confirmed that all HEART-enhanced versions significantly outperformed their respective base models ($P < 0.001$ for all comparisons), as visualized in Fig. 8B. Specifically, the average overall quality score increased from 3.57 (SD 0.72) to 4.20 (SD 0.41) for Llama-3.3 (Cohen's $d = 1.075$), from 3.92 (SD 0.70) to 4.38 (SD 0.42) for Claude-3.5 (Cohen's $d = 0.797$), from 3.91 (SD 0.73) to 4.32 (SD 0.39) for ChatGPT-4o (Cohen's $d = 0.701$), and from 4.03 (SD 0.69) to 4.38 (SD 0.41) for Gemini-2.5 (Cohen's $d = 0.617$) (all $P < 0.001$). Detailed mean scores and significance values are reported in Supplementary Table 10. While Gemini-2.5 showed slightly stronger baseline performance compared to other base models, the differences were not statistically significant. Among the enhanced models, Claude-3.5 + HEART and Gemini-2.5 + HEART achieved the highest overall quality scores, with no significant difference between them ($P > 0.05$), indicating comparable top-tier performance.

Discussion

Principal findings

Our findings indicate that the systematic integration of evidence-based medical knowledge with advanced language models can effectively tackle the critical challenges of accuracy, reliability, and safety in healthcare information generated by LLMs and RAG. The HEART framework offers a comprehensive and scalable solution for evidence-based hypertension education and clinical support, establishing a replicable model for domain-specific medical artificial intelligence (AI) applications.

Comparison with prior work

Beyond the well-established role in prognosis prediction, AI is increasingly being leveraged for patient education

Table 2 Comparative Five Dimensions by Clinicians for Base and HEART-Enhanced Versions of Llama-3.3, Claude-3.5, Gemini-2.5, and ChatGPT-4o on 50 Translated Hypertension-Related Questions

Mean Score (SD)	Llama-3.3			Claude-3.5			Gemini-2.5			ChatGPT-4o		
	Without HEART	With HEART	P value	Without HEART	With HEART	P value	Without HEART	With HEART	P value	Without HEART	With HEART	P value
Accuracy	3.46 (0.92)	4.19 (0.73)	<0.001	3.87 (0.91)	4.36 (0.53)	<0.001	4.12 (0.81)	4.37 (0.55)	0.011	3.99 (0.89)	4.30 (0.50)	0.005
Completeness	3.57 (0.83)	4.18 (0.72)	<0.001	3.94 (0.82)	4.41 (0.55)	<0.001	4.01 (0.76)	4.43 (0.59)	<0.001	4.01 (0.80)	4.37 (0.60)	<0.001
Consistency	3.68 (0.94)	4.32 (0.51)	<0.001	4.01 (0.81)	4.37 (0.54)	<0.001	4.00 (0.78)	4.32 (0.53)	<0.001	3.75 (0.85)	4.39 (0.55)	<0.001
Security	3.85 (0.90)	4.31 (0.64)	<0.001	4.01 (0.88)	4.39 (0.52)	<0.001	4.13 (0.80)	4.43 (0.50)	0.001	4.06 (0.87)	4.36 (0.50)	0.006
Robustness	3.28 (1.36)	4.04 (0.90)	<0.001	3.74 (0.99)	4.37 (0.56)	<0.001	3.91 (0.88)	4.35 (0.58)	<0.001	3.73 (0.90)	4.2 (0.72)	<0.001
Overall Quality	3.57 (0.72)	4.2 (0.41)	<0.001	3.92 (0.70)	4.38 (0.42)	<0.001	4.03 (0.69)	4.38 (0.41)	<0.001	3.91 (0.73)	4.32 (0.39)	<0.001

[24–31]. Previous investigations have preliminarily explored the application of LLMs in hypertension-related patient education, revealing both promising potential and significant limitations in delivering clinically accurate information on hypertension prevention and blood pressure management [32–35]. A systematic evaluation of 100 hypertension-specific queries demonstrated that ChatGPT generated clinically inappropriate responses in 7.5% of cases, with an additional 7% non-reproducibility rate, highlighting concerns regarding consistency and reliability [36]. Subsequent comparative analysis by Lee et al. assessed ChatGPT and Gemini performance across 52 hypertension consultation scenarios, reporting full correctness rates of merely 51.4% and 59.2%, respectively [37]. Our previous comprehensive evaluation revealed that ChatGPT-4o achieved satisfactory ratings (“good” or superior) in only 77% of responses for clarity, 73% for completeness, 73% for clinical utility, and a concerning 35% for conciseness when addressing common hypertension inquiries [20]. These convergent findings illuminate the dichotomous nature of current LLMs capabilities—while demonstrating capacity for generating contextually relevant content, their performance exhibits substantial heterogeneity across critical evaluation dimensions, particularly regarding consistency, completeness, and clinical reliability. Moreover, LLMs models remain susceptible to a critical limitation known as hallucination—a phenomenon wherein models generate plausible yet factually incorrect information that lacks correspondence to underlying training data or verifiable knowledge sources [38]. This propensity for fabricating content poses particularly acute risks in high-stakes domains such as healthcare, where inaccurate information can have severe consequences for patient safety and clinical decision-making processes [39].

Contemporary research has investigated diverse methodological approaches to enhance LLMs reliability and clinical applicability within healthcare domains, with three predominant strategies emerging: prompt engineering, supervised fine-tuning, and RAG [40]. Prompt engineering involves the strategic formulation of input instructions to optimize model output accuracy and clinical relevance [41]. While this approach offers implementation flexibility and cost-effectiveness, its efficacy demonstrates considerable variability and remains constrained by the model’s pre-existing knowledge boundaries, precluding the incorporation of novel or highly specialized clinical information beyond the original training corpus. Conversely, supervised fine-tuning enables LLMs to assimilate domain-specific patterns through targeted training on curated medical datasets [40]. Our previous investigation demonstrated that fine-tuning the GLM-4-9B model with hypertension-specific content yielded responses rated as “satisfactory” or superior



Fig. 6 Average scores for security and score distributions of responses from large language models

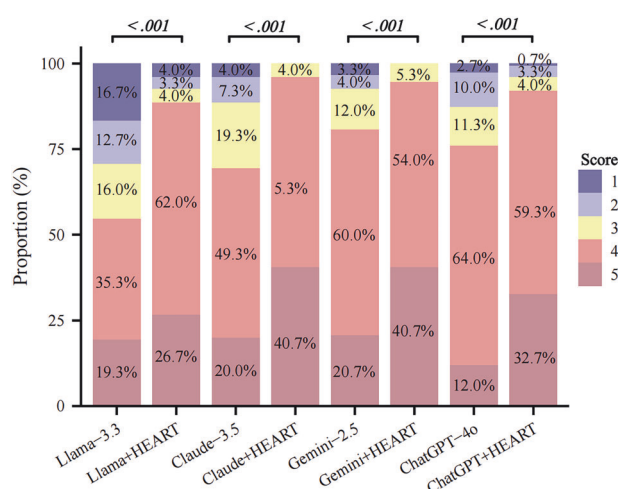


Fig. 7 Robustness Score Distributions of Responses from Large Language Models. HEART Hypertension Enhancing Answer Retrieval Tool

across multiple dimensions: clarity (80%), completeness (88%), conciseness (88%), and utility (83%) [20]. However, fine-tuning methodologies present substantial limitations, including intensive computational resource requirements and the necessity for periodic model retraining to incorporate evolving clinical guidelines, thereby constraining scalability and real-time adaptability in dynamic healthcare environments.

In contrast, RAG technology offers a more sustainable and adaptive solution by dynamically integrating external knowledge repositories with LLM capabilities during inference [42]. This architectural approach enables real-time access to current medical evidence while maintaining system adaptability without requiring iterative model retraining. Several recent investigations have validated RAG effectiveness across diverse clinical applications. The RISE

framework for diabetes education demonstrated that structured knowledge integration with prompt refinement significantly enhanced LLM response accuracy and clinical comprehensibility [43]. Similarly, DocOA—a specialized osteoarthritis management assistant—utilized RAG architecture with instructional prompting to outperform general-purpose models, including ChatGPT-3.5 and ChatGPT-4 in delivering personalized, evidence-based care recommendations [44]. In the nutrition informatics, RAG integration with the iDISC2.0 knowledge base achieved remarkable accuracy rates of 99% for binary classification tasks and 95% for multiple-choice assessments, substantially exceeding GPT-4o's performance of 62% and 52% on identical benchmarks [45]. These investigations collectively demonstrate RAG's superior capability in enhancing factual accuracy, clinical consistency, and domain-specific robustness. However, no RAG-based framework has been specifically developed or clinically validated for hypertension education and consultation applications.

Our investigation addresses this critical knowledge gap through the introduction of the HEART framework—the novel domain-specific RAG architecture designed exclusively for hypertension-related clinical question answering. Cohen's *d* values greater than 0.5 are generally regarded as educationally significant within the context of medical education research. The effect sizes observed in our study, which ranged from 0.617 to 1.075, indicate meaningful practical implications. Furthermore, the substantial effect sizes associated with overall quality improvements suggest that HEART enhancement generates responses that closely meet the quality standards typically expected from specialized clinical consultations. Through systematic integration of the meticulously curated HKBS2.0 knowledge repository with real-time retrieval and generation mechanisms, the HEART framework delivers responses that demonstrate not only enhanced accuracy and completeness but also consistent alignment with contemporary clinical practice guidelines. This work contributes substantial evidence supporting the strategic integration of structured medical knowledge into generative AI architectures for healthcare applications. Beyond improving individual performance metrics, the HEART framework demonstrates how retrieval-based augmentation can effectively bridge the critical gap between general-purpose language model capabilities and the stringent requirements of clinical reliability, thereby substantially improving the safety profile and trustworthiness of AI-driven healthcare communication systems. These findings suggest that the HEART framework methodology may serve as a foundational reference for developing analogous domain-specific educational and decision-support tools across other chronic disease management areas, potentially catalyzing broader adoption of evidence-based AI solutions in clinical practice.

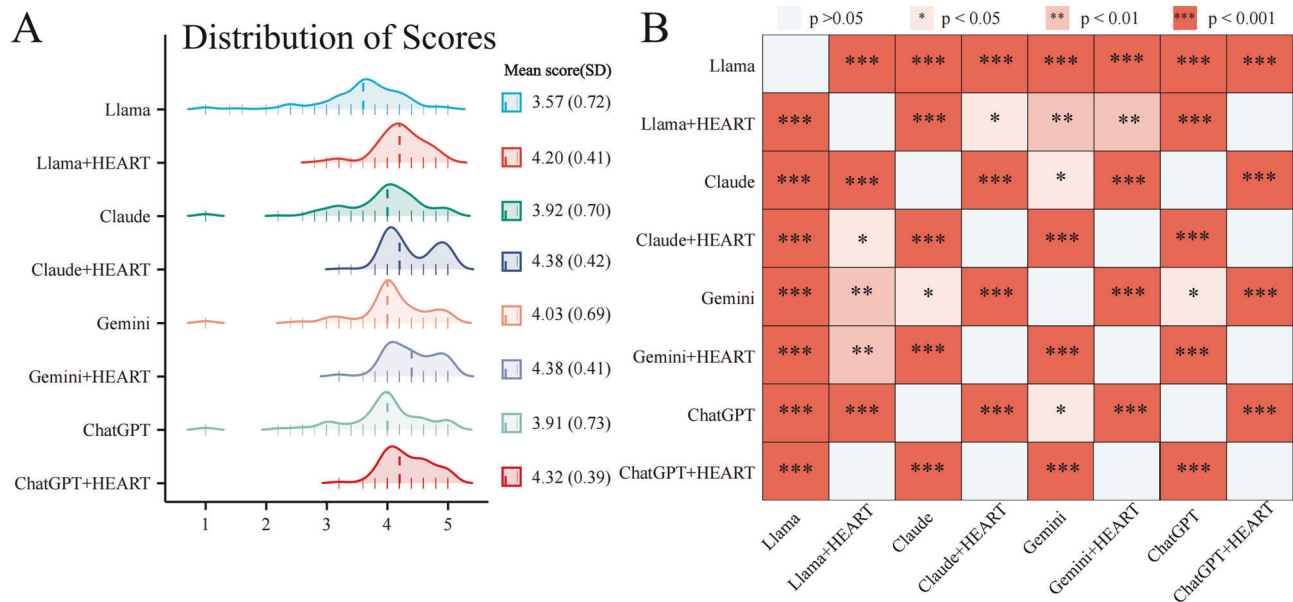


Fig. 8 Overall Quality Comparison of Base and HEART-Enhanced Models. **A** Distribution of average overall quality scores across eight models (four base LLMs and their HEART-enhanced versions), calculated as the mean of five evaluation dimensions. **B** Pairwise statistical comparisons of overall quality scores between models, presented

as a heatmap of P values. All HEART-enhanced models significantly outperformed their base counterparts ($P < 0.001$). *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$, LLM0 large language model, HEART0 Hypertension Enhancing Answer Retrieval Tool

Strengths and limitations

This study offers several notable strengths. First, the HEART framework represents the novel application of a RAG-based framework tailored specifically for hypertension education, addressing an important gap in domain-specific LLM-assisted patient communication. The integration of a rigorously curated Hypertension Knowledge Supplement Base ensures that the system's responses are grounded in authoritative and up-to-date clinical guidelines, thereby enhancing both the accuracy and clinical relevance of the generated answers. Additionally, the evaluation employed a comprehensive set of quality dimensions—including accuracy, completeness, consistency, security (likelihood of harm), and robustness—providing a multidimensional assessment of the system's performance.

Despite these strengths, several limitations should be acknowledged. First, while the HEART framework demonstrated significant improvements over base LLMs outputs, its performance remains contingent upon the quality and completeness of the underlying knowledge base; gaps or outdated information in the knowledge supplement may still affect response quality. Second, the evaluation focused primarily on technical and expert-assessed metrics without incorporating direct patient feedback, which is crucial to understanding real-world usability and patient satisfaction. Third, the system's reliance on retrieval mechanisms may introduce latency or computational overhead, potentially affecting

scalability in high-demand clinical settings. Fourth, the exclusive evaluation of the model using Chinese data derived from Chinese guidelines. While this approach has proven effective within the specific context, it raises questions about the model's adaptability to other linguistic and cultural settings. Therefore, further validation in different languages and cultural contexts will be necessary in future studies to assess the model's broader applicability.

Future work

While this study offers valuable insights into the application of the HEART framework, several areas warrant further investigation. First, ensuring the sustainability and continuous updating of RAG-based Q&A pairs represents a critical direction for future work. As new clinical guidelines and evidence emerge, the knowledge base must be systematically updated to preserve its accuracy and relevance. Moreover, implementing robust quality control mechanisms for Q&A pairs will be essential to maintain the trustworthiness and clinical validity of the information provided. Further research is also necessary to validate the HEART framework across diverse regions and healthcare systems, particularly in countries with varying medical guidelines and cultural contexts. Such studies will help evaluate the framework's adaptability and its capacity to deliver reliable hypertension education in different environments. Additionally, conducting a patient-centered validation study will be crucial to assess the system's real-world usability, with a

focus on key metrics such as patient satisfaction, readability, and perceived trustworthiness of the content [22]. Lastly, a more comprehensive analysis of the model's performance in handling ambiguous or uncertain queries is needed. Investigating how the model responds when information is limited, and how effectively it adheres to safety protocols—such as clearly stating “I do not know”—will provide valuable insights into the framework's reliability and safety. Addressing these future research directions will not only enhance the applicability and safety of the HEART framework but also contribute to its long-term sustainability and real-world impact.

Conclusion

This study demonstrates that integrating a high-quality, domain-specific hypertension knowledge base with RAG and LLMs can effectively address key limitations of standalone LLMs. The HEART framework significantly enhances the accuracy, completeness, and clinical relevance of hypertension-related information retrieval while substantially reducing the propagation of medical misinformation and improving the overall trustworthiness of AI-generated healthcare content. Through enabling dynamic access to contemporary, evidence-based medical knowledge, the HEART framework provides a scalable and clinically viable solution for patient education initiatives and chronic disease management protocols.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Author contributions JHW and JW conceived and performed the study. YJW, YJL, and SYC performed the clinical study and wrote the manuscript. JW, MLH, WPT, RJH and ZYY. performed data collection and analysis. All authors contributed to the article and approved the submitted version.

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Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

Ethics approval and consent to participate Ethical approval for this study was granted by the Ethics Committee of the First Affiliated Hospital of Bengbu Medical University (Approval No. [2024]KY066).

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